

CS201 Midterm 2

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Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures including arrays, (iv) you can use any function given in the data structure, (v) you can not modify the data structure contents.

I. QUESTION (QUEUE) (16 PTS)

Write the method for array implementation

```
void copyPaste(Queue src, int index)
```

which copies all the elements of the src queue and inserts to the queue at position index. You are not allowed to use enqueue, dequeue, isEmpty functions. You can assume the destination queue has enough space for insertion. Your method should run in $O(N)$ time. Hint: Start by counting the number of positions to shift for opening up space for the elements of src.

II. QUESTION (QUEUE) (15 PTS)

Write another constructor method

```
void Queue(Queue[] list)
```

which constructs a new list based queue by concatenating all elements in the list of queues in order. The elements from queues should be recreated (not copied from the queues). You are not allowed to use enqueue, dequeue, isEmpty functions. You should solve the question for list implementation.

III. QUESTION (BINARY SEARCH TREE) (12 PTS)

Write the recursive method

```
void accumulateLeafNodes(Queue queue)
```

in TreeNode class which accumulates the contents (integer as data field) of all leaf nodes in queue.

For queue, you are only allowed to use enqueue function. You should use array implementation for the queue in this question.

IV. QUESTION (BINARY SEARCH TREE) (21 PTS)

Write a non-recursive method

```
double simulateSearch(int N)
```

in Tree class, which first finds the minimum (A) and maximum (B) elements in the tree. The method will then randomly search a number between [A, B] N times and returns the average number of nodes visited in this search. You are not allowed to use any tree methods.

V. QUESTION (HASHING) (17 PTS)

Write a static method in Hash class

```
int[] sortByHashing(int[] array)
```

that takes an array of integers as a parameter (which contains distinct numbers less than $2N$, where N is the number of elements in the array) and returns the sorted version of that array. Your method should run in $O(N)$ time. Do not use any external data structures or arrays except the resulting array and hash table. Hint: Find the maximum number in the array, the sorted array should contain only numbers less than that.

VI. QUESTION (HASHING) (19 PTS)

Write the static method

```
int[] intersection(int[] list1,
                  int[] list2)
```

to find the intersection of the elements in two arrays and return a new array. Your method should run in $O(N)$ time, where N is the number of elements in the arrays. Do not use any external data structures or arrays except the resulting array and hash table.

The intersection array should contain only that many items not more not less. Hint: How can you search an element from the first list in the second list in $\mathcal{O}(1)$?